

Solutions

to Individual Competition Problems

Problem I-1

Determine all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y) + 2y) = f(xy) + xf(y) + f(f(y))$$

holds for all real numbers x and y .

(proposed by Patrik Bak, Slovakia)

Answer. The functional equation has two solutions, $f(x) \equiv 0$ and $f(x) \equiv 2x$.

Solution. Setting $x = 0$ and $y = 0$ in the functional equation yields $f(f(0)) = 0$. So there is at least one zero point of f . Let a be any of them. Setting $y = a$ gives us $f(2a) = f(ax) + f(0)$. If $a \neq 0$, then f is a constant function and we know that $f(a) = 0$, so it is a zero function, which is indeed a solution.

It remains to investigate the case where 0 is the only zero point of f , i.e. $f(a) = 0$ if and only if $a = 0$. Furthermore, taking $x = 0$ in the functional equation we obtain

$$f(2y) = f(f(y)). \quad (1)$$

If we prove an injectivity of f , the previous identity yields $f(y) = 2y$, what is the second solution, as we can easily check.

Now we prove the injectivity of f . Firstly, let us examine the set of the fixed points of f . This set is non-empty because 0 is one of its points. Assume that p is any of the fixed points, i.e. $f(p) = p$. Setting $x = -1$, $y = p$ in the functional equation gives

$$p = f(-f(p) + 2p) = f(-p) - f(p) + f(f(p)) = f(-p).$$

Now we set $x = 1$, $y = -p$ in the functional equation and we obtain using proved $f(-p) = p$

$$p = f(f(-p) - 2p) = f(-p) + f(-p) + f(f(-p)) = 3p.$$

This yields that $p = 0$ is the only fixed point of f .

Secondly, we choose x so that $xf(y) + 2y = xy$, which yields $x = 2y/(y - f(y))$. This can be done for each $y \neq 0$, since 0 is the only fixed point of f . This substitution gives us

$$f(f(y)) = \frac{2yf(y)}{f(y) - y} = \frac{2f^2(y)}{f(y) - y} - 2f(y).$$

In order to finish the proof of injectivity let us assume that non-zero real numbers a, b satisfy $f(a) = f(b)$. We have already proved that $f(a) = f(b) \neq 0$. The previous identity yields

$$\frac{2f^2(a)}{f(a) - a} - 2f(a) = f(f(a)) = f(f(b)) = \frac{2f^2(b)}{f(b) - b} - 2f(b)$$

and it follows that $a = b$. The proof of the injectivity is thereby finished.

Remark. We present an alternative way of proving that $f(p) = p$ implies $p = 0$.

Assume that $f(p) = p$. Then $y = p$ in (1) means $f(2p) = p$ and afterwards $y = 2p$ means $f(4p) = p$. Taking $x = 2$ and $y = p$ in the original equation then gives us $p = 0$, which works. Therefore the only fixed point of f is 0.

Solution by Jozef Fülöp awarded by prize of the dean of the FMF, Charles University, Prague.

We set $x = 1$, $y = 0$ in the original equation and we obtain

$$f(f(0)) = f(0) + f(0) + f(f(0))$$

which implies $f(0) = 0$. This identity yields after setting $x = 0$ in the original equation

$$f(2y) = f(f(y)). \quad (2)$$

Let us assume that there exists $t_0 \neq 0$ such that $f(t_0) = 0$. For every real number t we obtain by substitution $x = t/t_0$, $y = t_0$ in the original equation

$$f(2t_0) = f(t) + \frac{t}{t_0} \cdot f(t_0) + f(f(y)) = f(t) + f(f(t_0)).$$

This equation with (2) gives us $f(t) = 0$ for each real number t , which is one of the solutions, as we can easily check.

Now we can assume that $f(y) \neq 0$ for every real number $y \neq 0$. For each real number t and $y \neq 0$ we obtain by putting $x = 2t/f(y)$ in the original equation

$$f(2t + 2y) = f\left(\frac{2ty}{f(y)}\right) + 2t + f(f(y)).$$

We prove that the function f is injective. If a, b are numbers from $\mathbb{R} \setminus \{0\}$ such that $f(a) = f(b)$ ($\neq 0$) then the substitutions $t = a, y = b$ or $t = b, y = a$ in the previous equation gives

$$\begin{aligned} f(2a + 2b) &= f\left(\frac{2ab}{f(b)}\right) + 2a + f(f(b)), \\ f(2b + 2a) &= f\left(\frac{2ba}{f(a)}\right) + 2b + f(f(a)). \end{aligned}$$

This two equations directly yields to $a = b$, which proves the injectivity.

The injectivity of the function f together with (2) lead to $f(y) = 2y$ what is the second solution, as we can check.

Problem I-2

Let $n \geq 3$ be an integer. We say that a vertex A_i ($1 \leq i \leq n$) of a convex polygon $A_1A_2 \dots A_n$ is *Bohemian* if its reflection with respect to the midpoint of the segment $A_{i-1}A_{i+1}$ (with $A_0 = A_n$ and $A_{n+1} = A_1$) lies inside or on the boundary of the polygon $A_1A_2 \dots A_n$. Determine the smallest possible number of Bohemian vertices a convex n -gon can have (depending on n).

(A convex polygon $A_1A_2 \dots A_n$ has n vertices with all inner angles smaller than 180° .)

(proposed by Dominik Burek, Poland)

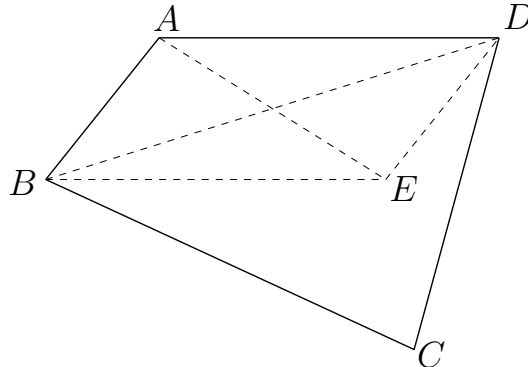
Answer. $n - 3$.

In the following we write for short ‘reflection of A in P ’ instead of ‘reflection of the vertex A with respect to the midpoint of the segment connecting the two neighbouring vertices of A in the polygon P ’.

Solution.

Lemma. *If $ABCD$ is a convex quadrilateral with $\angle BAD + \angle CBA \geq \pi$ and $\angle BAD + \angle ADC \geq \pi$ then A is a Bohemian vertex of $ABCD$.*

Proof. Let E be the reflection of A in $ABCD$. It is clearly seen that E belongs to the halfplanes containing C determined by lines AB and AD . Since $\angle BAD + \angle CBA \geq \pi$ and $\angle BAD + \angle EBA = \pi$, point E belongs to the (closed) halfplane containing points A, D determined by the line BC . Analogously, using the assumption $\angle BAD + \angle ADC \geq \pi$ we infer that E belongs to the closed halfplane containing points A, B determined by the line CD .



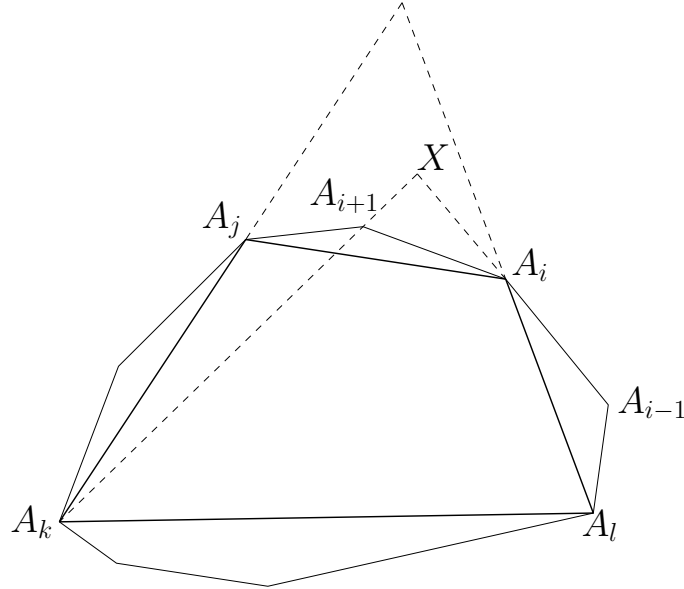
Therefore E lies inside or on the boundary of $ABCD$. Thus A is Bohemian. □

Consider a convex n -gon $A_1 A_2 \dots A_n$. Choose any four vertices A_i, A_j, A_k, A_l with $i < j < k < l$ as in the picture below. Consider quadrilateral $A_i A_j A_k A_l$. It is clear that one of the points A_i, A_j, A_k, A_l satisfies assumption of the lemma, let's say this point is A_i . We claim that A_i satisfies the assumption of the lemma in quadrilateral $A_{i-1} A_i A_{i+1} A_k$. Observe that the point $X := A_k A_{i+1} \cap A_i A_{i-1}$ lies in the triangle bounded by lines $A_k A_j$, $A_j A_i$ and $A_i A_l$. So

$$\angle A_k A_{i+1} A_i + \angle A_{i+1} A_i A_{i-1} = \pi + \angle A_k X A_i \geq \pi.$$

(Note: it may happen that X does not exist. It happens iff $j = i + 1$, $l = i - 1$ and $A_k A_j \parallel A_l A_i$. In that case $\angle A_k A_{i+1} A_i + \angle A_{i+1} A_i A_{i-1} = \pi$.)

Analogously $\angle A_{i+1} A_i A_{i-1} + \angle A_i A_{i-1} A_k \geq \pi$. Using lemma we conclude that A_i is a Bohemian vertex of quadrilateral $A_{i-1} A_i A_{i+1} A_k$. This implies that A_i is a Bohemian vertex of $A_1 A_2 \dots A_n$ since the quadrilateral $A_{i-1} A_i A_{i+1} A_k$ is a subset of the n -gon and the reflexion point is the same.



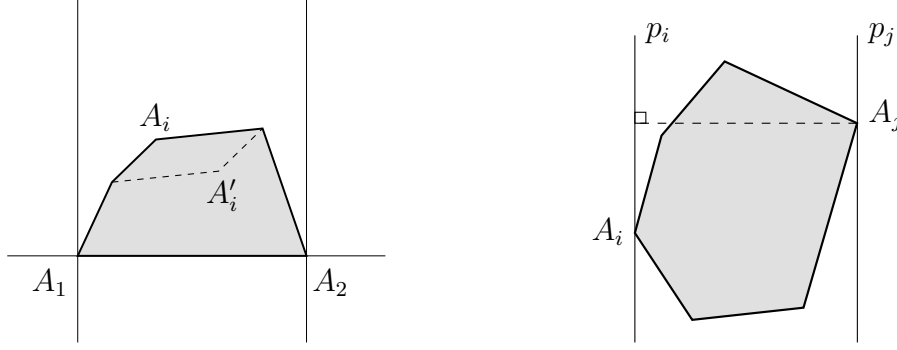
Therefore, amongst any four vertices of a convex n -gon there exists a Bohemian vertex. So, every n -gon has at least $n - 3$ Bohemian vertices.

An example of a convex n -gon with exactly $n - 3$ Bohemian vertices is the following: take any kite $A_1 A_2 A_3 A_4$ with $A_4 A_1 = A_1 A_2 < A_2 A_3 = A_3 A_4$ and place points $A_5, \dots, A_n$ very close to A_1 . Then A_2, A_3, A_4 are not Bohemian vertices of $A_1 A_2 \dots A_n$.

Solution 2. We present a sketch of an alternative proof of the fact that the every n -gon has at least $n - 3$ Bohemian vertices.

Observation 1. Let us place the polygon into a coordinate system in such a way that $A_1 = [0, 0]$, $A_2 = [a, 0]$, $a > 0$ and the second coordinates of all the remaining vertices are positive. If all the remaining vertices $A_2, \dots, A_n$ have their first coordinates between 0 and a (see picture below), it is easy to see that the only vertices that could be non-Bohemian are A_1, A_2 , and the

point with the strictly largest second coordinate (if such a vertex exists). So, in this case, there exist at least $n - 3$ Bohemian vertices.



Observation 2. An affine transformation does not change anything, so the statement is proved for all polygons that lie between two parallel lines that go through two adjacent vertices, i.e., whenever there are two adjacent vertices with sum of their angles at most 180° .

Consider now any polygon $P = A_1A_2, \dots, A_n$.

Observation 3. If there are two (non-adjacent) vertices A_i, A_j and two parallel lines p_i, p_j with $A_i \in p_i, A_j \in p_j$ such that the whole polygon lies between p_i and p_j , then the diagonal A_iA_j splits P into two polygons of the type considered in Observation 2. By Observations 1 and 2, these two polygons have at most 4 non-Bohemian points together, A_i, A_j , and two more.

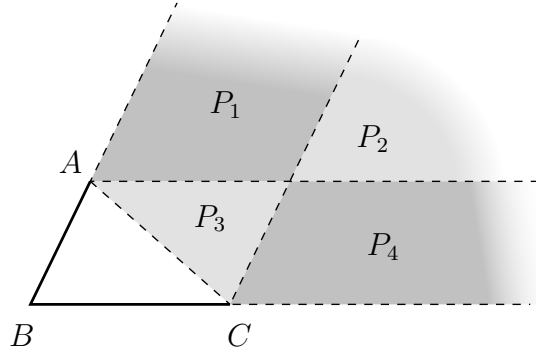
Observation 4. For any vertex A_i there exist a vertex A_j and two parallel lines p_i and p_j with $A_i \in p_i, A_j \in p_j$ such that the whole polygon lies between them. In fact, take a line p_i such that $p_i \cap P = \{A_i\}$, then A_j is the vertex with maximal distance from p_i , if there are two such vertices, change the direction of p_i slightly to obtain a unique A_j .

Let $n = 4$. Then there exist two adjacent vertices with sum of their angles not larger than 180° , so, by Observation 2, any quadrilateral has at most 3 non-Bohemian vertices.

Let $n \geq 5$. By Observations 3 and 4, there exist at most 4 non-Bohemian vertices. So, at least one vertex is Bohemian, denote it by A_i . Then, by Observations 3 and 4, all the non-Bohemian vertices are contained in the quadruple A_i, A_j and some other two vertices. Since A_i is Bohemian, there are at most three non-Bohemian vertices and the proof is complete.

Solution 3. We prove by induction that every n -gon has at least $n - 3$ Bohemian vertices.

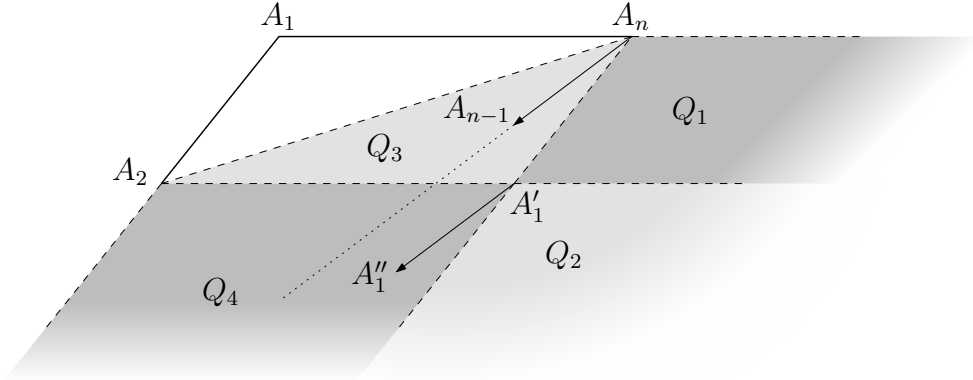
Step 1, $n = 4$. We show that every quadrilateral $ABCD$ has at least one Bohemian vertex. We consider a triangle ABC . Then D has to be in one of the areas P_1, P_2, P_3, P_4 (see picture below), otherwise, $ABCD$ would not be a convex quadrilateral. If D were in P_2 , then B would be Bohemian. If D were in P_3 , it would be Bohemian. If D were in P_1 , then A would be Bohemian and similarly if D were in P_4 , then C would be Bohemian (the last two cases are not immediate but easy to prove).



Induction step. Consider an n -gon $P = A_1A_2 \dots A_n$ with $n \geq 5$. Let P' be an $(n-1)$ -gon obtained from P by omitting one vertex different from A_1 . Let A'_1 be the reflexion of A_1 in P and A''_1 the reflexion of A_1 in P' . We show the following statement

if A_1 is non-Bohemian in P , then it is non-Bohemian also in P' . (S)

This statment is obvious if the omitted vertex is not adjacent to A_1 (since in this case $A'_1 = A''_1$ and $P' \subset P$). So, let the omitted vertex be A_n (the other neighbour A_2 can be done in the same way) and let us assume for contradiction that $A'_1 \notin P$ and $A''_1 \in P'$. Let us observe that vectors A_nA_{n-1} and $A'_1A''_1$ are equal. Let us discuss the possible position of A_{n-1} . If $A_{n-1} \in Q_3 \cup Q_4$ (as in the picture below) then A'_1 lies below the line A_nA_{n-1} while the whole polygon P lies above this line, contradiction with $A''_1 \in P' \subset P$. If $A_{n-1} \in Q_2$, then $A'_1 \in \triangle A_2A_nA_{n-1} \subset P$, contradiction. If $A_{n-1} \in Q_1$, then the new reflexion $A''_1 \in Q_2$ and $A'_1 \in \triangle A_2A_nA''_1$ and $\triangle A_2A_nA''_1 \subset P'$ since $A''_1 \in P'$. Therefore $A'_1 \in P' \subset P$, contradiction. Statement (S) is proved.



Since (by the induction hypothesis) there are at most 3 non-Bohemian vertices in P' , there are at most 4 non-Bohemian vertices in P (the three and the omitted one). Since $n \geq 5$ there is at least one Bohemian vertex in P . Assume now that P' is obtained form P by omitting a Bohemian vertex. Since there are at most 3 non-Bohemian vertices in P' , there are at most 3 non-Bohemian vertices in P and the proof is complete.

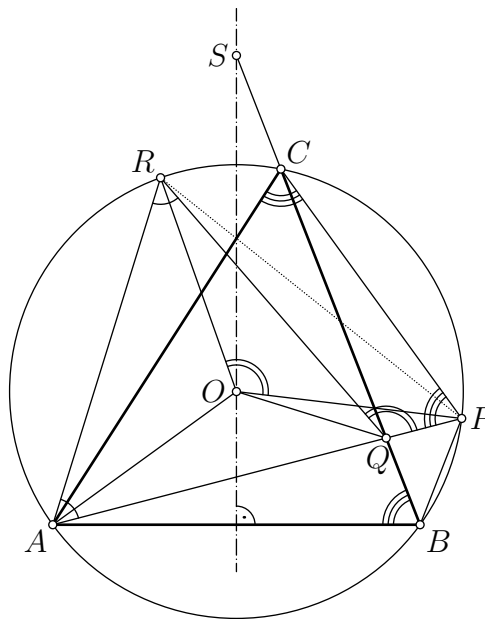
Problem I-3

Let ABC be an acute-angled triangle with $AC > BC$ and circumcircle ω . Suppose that P is a point on ω such that $AP = AC$ and that P is an interior point of the shorter arc BC of ω .

Let Q be the point of intersection of the lines AP and BC . Furthermore, suppose that R is a point on ω such that $QA = QR$ and that R is an interior point of the shorter arc AC of ω . Finally, let S be the point of intersection of the line BC with the perpendicular bisector of the side AB . Prove that the points P , Q , R , and S are concyclic.

(proposed by Patrik Bak, Slovakia)

Solution. Let us denote O the center of the circle ω and $\varphi = \angle PAR$. Since the triangle QAR is isosceles, we have $\angle ARQ = \varphi$ and $\angle PQR = 2\varphi$. The central angle theorem (applying to the chord PR) also yields $\angle POR = 2\varphi$. Thus the points P , Q , O and R are concyclic.



Further, let us denote $\beta = \angle ABC$. Since $AP = AC$, we have $\angle ACP = \angle APC = \beta$ and thus (by the central angle theorem) $\angle AOP = 2\beta$, which gives

$$\angle PAO = \angle APO = 90^\circ - \beta = \angle OPQ.$$

Since $\angle ABS = \beta$, we furthermore have $\angle OSB = \angle OSQ = 90^\circ - \beta$, which concludes $\angle OPQ = \angle OSB$ and therefore also the points P , Q , O and S are concyclic.

From both paragraphs above it immediately follows the requested claim, i.e. the points P , Q , R , S are concyclic, and the proof is done.

Problem I-4

Determine the smallest positive integer n for which the following statement holds true: From any n consecutive integers one can select a non-empty set of consecutive integers such that their sum is divisible by 2019.

(proposed by Kartal Nagy, Hungary)

Answer. $n = 340$.

Solution The prime factorization of 2019 is $3 \cdot 673$. Let $p = 673$.

For each integer k , color the three numbers $kp - 1, kp, kp + 1$ red, and the six numbers $kp + \frac{p-5}{2}, kp + \frac{p-3}{2}, kp + \frac{p-1}{2}, kp + \frac{p+1}{2}, kp + \frac{p+3}{2}, kp + \frac{p+5}{2}$ blue. Now the integers are colored periodically. In a period of length $p = 673$, there are 3 red integers, then 332 uncolored integers, then 6 blue integers and finally 332 uncolored integers.

The sum of the integers in a red interval is $3kp = 2019 \cdot k$, and the sum of the integers in a blue interval is $6(kp + \frac{p}{2}) = 2019 \cdot (2k + 1)$. So if there is a colored interval (we mean a maximal one throughout) in the given n consecutive integers, one can choose it. It is easy to see, that among any $340 = 332 + (6 - 1) + (3 - 1) + 1$ consecutive integers, there must be a colored interval. Thus the smallest n (that we look for) satisfies $n \leq 340$.

Now we will show that it is not possible to choose consecutive integers in the desired way from the set $A = \{335, 336, \dots, 673\}$. ($|A| = 339$ and thus $n \geq 340$.) Assume that there exists $\{a, a + 1, \dots, b\} \subseteq A$ such that

$$2019 \mid a + (a + 1) + \dots + b = \frac{(b - a + 1)(a + b)}{2}.$$

That means either $673 \mid b - a + 1$, or $673 \mid a + b$. Since

$$0 < 1 \leq b - a + 1 \leq 339 < 673,$$

673 must divide $a + b$. Taking into account that

$$671 = 335 + 336 \leq a + b \leq 673 + 673 = 2 \cdot 673,$$

we conclude that $a + b$ must be 673 or $2 \cdot 673$. It means either $a = 335$ and $b = 338$, or $a = 336$ and $b = 337$, or $a = b = 673$. But $2019 \nmid 335 + 336 + 337 + 338 = 1346$, $2019 \nmid 336 + 337 = 673$ and $2019 \nmid 673$, a contradiction.

Comment. The same proof works for every odd number $m = p \cdot q$, where p is a ‘big’ prime divisor of m . We need that $p > \sqrt{3m}$. Then the answer is $n = \frac{p+3q}{2} - 1$.